EXPERIMENT NO: 18

Experiment to understand KNN algorithm for a given dataset

PROGRAM:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Iris.csv')
print(df.info())
print(df.variety.value_counts())
print(df.head())
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,random_state=42)
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
from sklearn.metrics import confusion_matrix
print(confusion_matrix(label,model_KNN.predict(features)))
from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

variety	count
Setosa	50
Versicolor	50
Virginica	50

dtype: int64

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

KNeighborsClassifier

KNeighborsClassifier()

0.9666666666666667
1.0

array([[50, 0, 0],
[0, 47, 3],
[0, 1, 49]])

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	50
Versicolor	0.98	0.94	0.96	50
Virginica	0.94	0.98	0.96	50
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150